

STEP Interface Functional Specification

Institute of Technical Education - CRM System

Version 1.0



Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	01/11/2023	The initial version of the Alumni Interface FS	AvePoint
0.2	08/11/2023	Add response flow in API diagram	AvePoint
0.3	14/11/2023	Add section 2.3 Assumption	AvePoint
1.0	15/11/2023	Signed-off version	AvePoint



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1. Introduction

1.1 Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS Skills Training Enhancement Portal (STEP) and Institute of Technical Education CRM System. This interface will be a 2-ways interface between two systems. The purpose of this interface is for ITE alumni to enjoy alumni discounts when applying for specific courses in STEP. It is required to check Alumni status in ITE CRM system in real-time.

1.2 Intended Audience

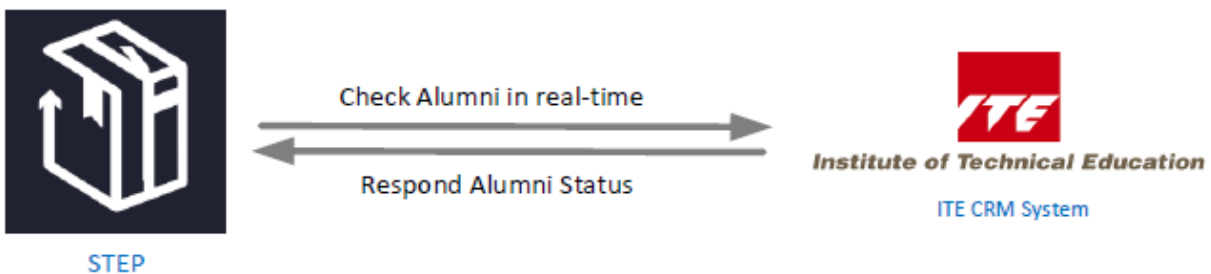
This technical document is used for system communication. Therefore, the intended audiences are the stakeholders of both STEP and ITE CRM.

2. Technical Overview

There are 2 roles involved in the interface diagram.

- **ITE CRM:** Institute of Technical Education CRM System
- **STEP:** Skills Training Enhancement Portal

2.1 System Flow



1. Upon student application, STEP will call ITE CRM API to check the applicant's alumni status.
2. ITE CRM responds applicant's alumni status (Yes/No) to STEP.

2.2 Interface Design Summary

Check ITE alumni status

Interface Function	Check alumni status
Hosting System	ITE CRM
Data Source System	ITE CRM
Data Destination System	Skills Training Enhancement Portal
API Name	CheckAlumniStatus
Mode of Interface	HTTP API
Method	POST
Content-Type	application/json
Frequency	Realtime
Encryption Method	HTTPS
Endpoint URL	<TBC>

STEP allows ITE to configure if alumni discount is available under service management. If available, the student who is an Alumni will get the corresponding discount configured by the IHL. For course setup details, please refer to STEP stage 1 functional specification: 6.3 Manage course. For discount setup details, please refer to STEP stage 1 functional specification: 5.8 Manage Discount.

Course

Student finance

Admin

Version 2

<

Course configuration

1 General information

2 Course details

3 Course fee

Course / Course configuration / Create course

4



Course fee

Preview

Course setup

Course type *
Academic Certificate

Billing by *
PDC/MC

Rate effective date base on *
Course intake date

Allow payment by *
PSEA x SFC x Credit/Debit Card x

Course rate set up by
Course rate table

Fee setup

Course

Specialist diploma in e-Business Managem...

Post Diploma Certificate

Post Diploma in MSc e-Business Manage...

Module (e-Business Management)

Statement of result

Module (Digital Development with ...)

None

Post Diploma Certificate

Specialist diploma in Digital Developme...

Module (Digital Development with ...)

Statement of result

Statement of participation

Course rate setup

Rate code
PTD-AC-BMO-NWSP

Rate version
Version 1

Rate type
Hourly rate per pax

Current rate
21.8

Funding

☒ SSG Cat A [View details](#)

Supplementary fee

☒ Miscellaneous fee [View details](#)

Discount

☒ Alumni discount [View details](#)

☐ General discount [View details](#)

Cancel

Save as draft

Back

Complete

4

2.3 Assumption

ITE Alumni interface is dependent on Data repository (Data lake) Interface for CET part, Data Repository Interface would include the correct data set to identify students who have graduated from STEP.

3. Interface Fields

3.1 Check ITE Alumni Information

Request parameters

STEP Field	CRM Field	Data Type	Length	Remarks
NRIC/FIN	NRIC	Char	9	Sample: G2398709K Checks with full NRIC

Response

STEP Field	CRM Field	Data Type	Length	Remarks
Alumni Status	alumni_status	Char	3	Sample: Yes/No

3 Error Handling

Please refer to the below table for response code that ITE CRM will return to STEP.

If the API throws an exception (other than response code 200), the applicant will not be recognized as an ITE alumni in STEP.

Response Code	Scenario	Type
200	Successful	Success
401	- BadAuthTicket - ExpiredAuthTicket - InsufficientAuthTicket - InvalidAuthTicket - InvalidUserAuth - MissingCrmAuthenticationToken - MissingCrmAuthenticationTokenOrganizationName - RequestIsNotAuthenticated	Client Error
403	- AccessDenied - AttributePermissionReadIsMissing - AttributePermissionUpdateIsMissingDuringUpdate - AttributePrivilegeCreateIsMissing - CannotActOnBehalfOfAnotherUser - CannotAddOrActOnBehalfAnotherUserPrivilege	Client Error
503	Service Unavailable Expect this status code when the web API service is not available.	Server Error